Clinton-Essex-Lake Champlain, NY 2014 QL2 LiDAR Project Report

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1. Summary / Scope

1.1. Summary

This report contains a summary of the Clinton-Essex-Lake Champlain, NY 2014 QL2 LiDAR acquisition task order, issued by the USGS National Geospatial Technical Operations Center, under their Spatial Products and Services Contract issued on September 19, 2014 and modified on December 19, 2014. The task order yielded a project area covering approximately 2,669 square miles over northeastern New York near Lake Champlain. The intent of this document is to only provide specific validation information for the LiDAR data acquisition/collection work completed as specified in the task order.

1.2. Scope

Aerial topographic LiDAR was acquired using state of the art technology along with the necessary surveyed ground control points (GCPs) and airborne GPS and inertial navigation systems. The aerial data collection was designed with the following specifications listed in Table 1 below.

Table 1. Originally Planned LiDAR Specifications

Average Point Density	Field of View	Minimum Side Overlap	RMSEz
2.0 pts / m^2	40°	10%	9.25 cm or better

1.3. Coverage

The LiDAR project boundary covers approximately 2,669 square miles over the square miles and encompasses parts of Clinton, Essex, and Franklin Counties in northeastern New York. The Base AOI is approximately 1,951 square miles. Data for the two optional areas of interest, Ausable River (289 square miles) and Saranac River (429 square miles), was also processed. These extents are shown in Figure 1 on the following page.

1.4. Duration

LiDAR data was acquired from November 2, 2014 to May 8, 2015 in 28 total lifts. See “Section: 2.6. Time Period” for more details.

1.5. Issues

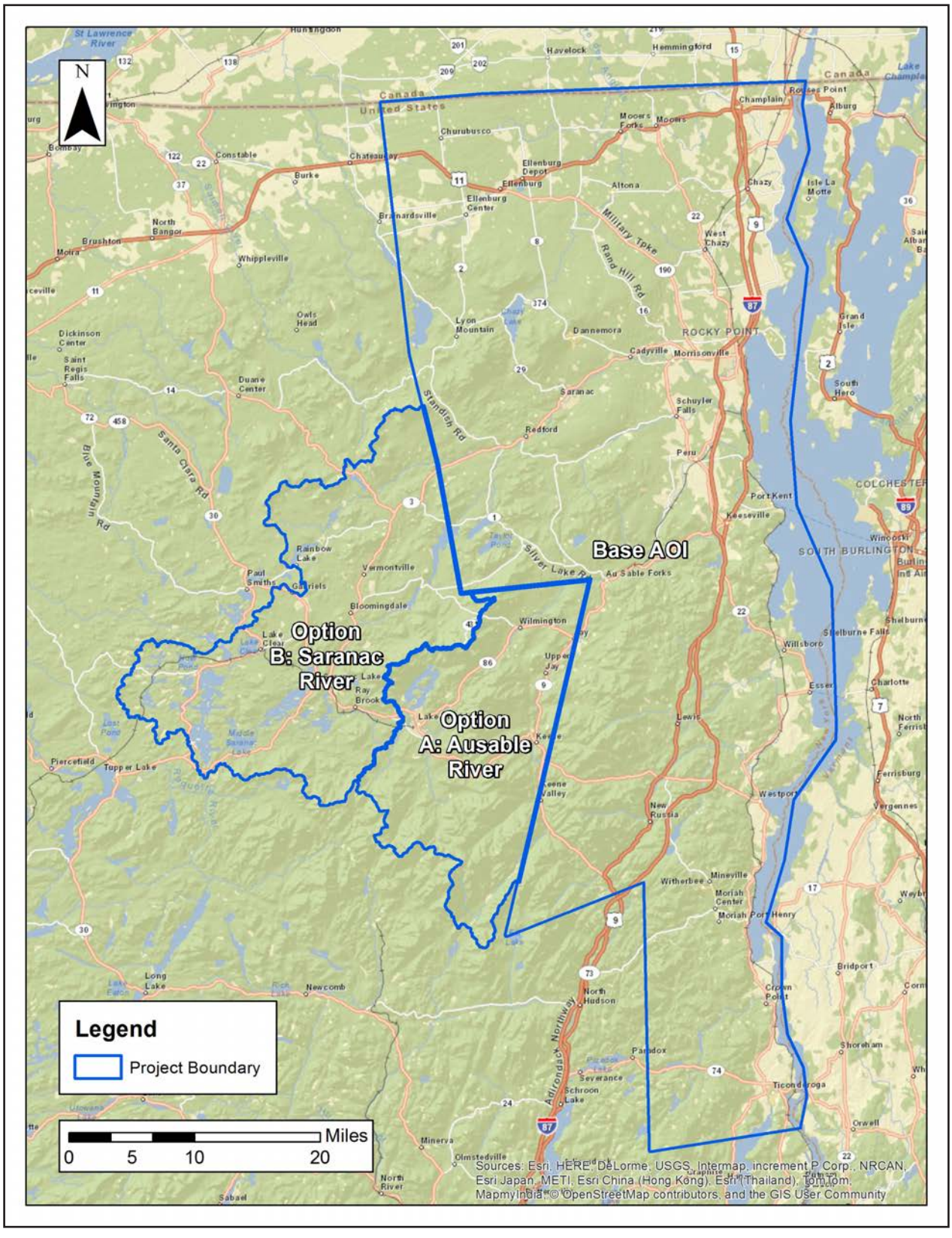
There were no issues with this project.

1.6. Deliverables

The following products were created for this project:

- Raw point cloud swath LAS in version 1.2 format
- Classified point cloud tiled LAS in version 1.2 format
- Hydro flattened raster DEM in ERDAS .IMG format
- Hydro flattened breaklines in ESRI shapefile format
- Ground control points in ESRI shapefile format
- Intensity Images in GeoTIFF format
- Flightline swaths in ESRI shapefile format
- Tile index in ESRI shapefile format
- 1-foot Contours in ESRI shapefile format
- Processing boundary in ESRI shapefile format
- Project, deliverable, and tile level metadata in XML format
- Accuracy Assessment in XLS format

Figure 1. Clinton-Essex-Lake Champlain NY 2014 LiDAR Project Boundary



2. Planning / Equipment

2.1. Flight Planning

Flight planning was based on the unique project requirements and characteristics of the project site. The basis of planning included: required accuracies, type of development, amount / type of vegetation within project area, required data posting, and potential altitude restrictions for flights in project vicinity. Please note that certain values in the table below are listed as "Variable" due to the various flight plans used, as described in "Section: 1.5. Issues" of this document.

Detailed project flight planning calculations were performed for the project using Leica MissionPro planning software.

2.2. LiDAR Sensor

Quantum Spatial utilized Leica ALS70 LiDAR sensors (Figure 2), serial number(s) 6121 and 7234 , during the project. The system is capable of collecting data at a maximum frequency of 500 kHz, which affords elevation data collection of up to 500,000 points per second. The system utilizes a Multi-Pulse in the Air option (MPIA). The sensor is also equipped with the ability to measure up to 4 returns per outgoing pulse from the laser and these come in the form of 1st, 2nd, 3rd and last returns. The intensity of the returns is also captured during aerial acquisition.

A brief summary of the aerial acquisition parameters for the project are shown in the LiDAR System Specifications in Table 2.

Table 2. Lidar System Specifications

		7121 Kucera 4B6	7121 Kucera BTV	7121 Kucera SP3	7121 Adirondack	7234 OptionA
Terrain and Aircraft Scanner	Flying Height (m)	701 - 1376	820 - 1376	655 - 1377	1483 - 2310	288 - 1775
	Recommended Ground Speed (kts)	150	150	150	160	105
Scanner	Field of View (deg)	39.0	40.0	40.0	28.0	40.0
	Scan Rate Setting Used (Hz)	41.5	41.0	34.6	52.2	27.3
Laser	Laser Pulse Rate Used (kHz)	194000	193400	193400	243200	153.8
	Multi Pulse in Air Mode	Disabled	Disabled	Disabled	Enabled	Disabled
Coverage	Full Swath Width (m)	974.88	1002.00	1002.13	1151.90	1292.09
	Line Spacing (m)	335.97	430.60	336.40	581.67	131.12
Point Spacing and Density	Maximum Point Spacing Along Track (m)	0.93	0.94	1.12	1.10	0.99
	Maximum Point Spacing Along Track (m)	0.93	0.94	1.12	0.79	0.99
	Average Point Density (pts / m^2)	2.58	2.50	2.50	2.57	2.20

Figure 2. Leica ALS70 LidAR Sensor



2.3. Aircraft

All flights for the project were accomplished through the use of a customized Cessna Caravan (single-turboprop), Tail Number N269JE. These aircraft provided an ideal, stable aerial base for LiDAR acquisition. These aerial platforms has relatively fast cruise speeds which are beneficial for project mobilization / demobilization while maintaining relatively slow stall speeds which proved ideal for collection of high-density, consistent data posting using a state-of-the-art Leica ALS70 LiDAR system. Some of the operating aircraft can be seen in Figure 3 below.

Figure 3. Some of Quantum Spatial's Planes



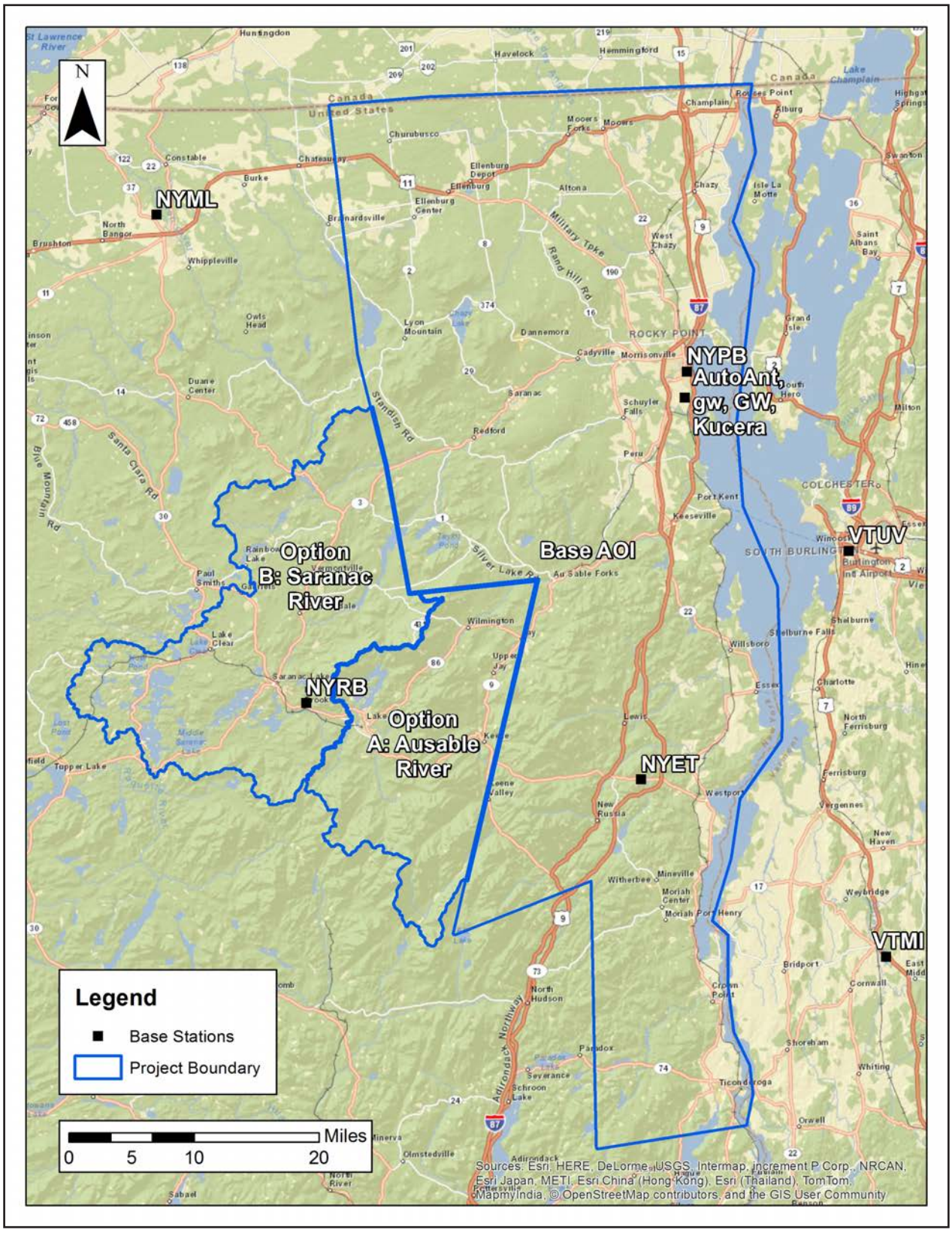
2.4. Base Station Information

GPS base stations were utilized during all phases of flight (Table 3). The base station locations were verified using NGS OPUS service and subsequent surveys. Base station locations are depicted in Figure 4. Data sheets, graphical depiction of base station locations or log sheets used during station occupation are available in Appendix A.

Table 3. Base Station Locations

Base Station	Latitude	Longitude	Ellipsoid Height (m)
NYET	44° 12′ 34.8508″	73° 32′ 25.85231″	174.246
NYML	44° 52′ 15.38494″	74° 17′ 107.34324″	195.213
NYPB	44° 40′ 50.58254″	73° 27′ 14.30528″	30.15
NYRB	44° 18′ 13.9228″	74° 4′ 42.07519″	466.846
VTMI	43° 59′ 55.02507″	73° 9′ 9.38024″	96.027
VTUV	44° 28′ 9.21279″	73° 11′ 52.37575″	112.41
AutoAnt, gw, GW, Kucera	44° 39′ 1.91729″	73° 27′ 30.704″	20.665

Figure 4. Base Station Locations



2.5. Time Period

Project specific flights were conducted over several months. Twenty-eight LiDAR sorties, or aircraft lifts were completed. Accomplished sorties are listed below.

LiDAR Sorties

- Apr 29, 2015-A (7121)
- Apr 29, 2015-B (7121)
- Apr 29, 2015-C (7121)
- Apr 29, 2015-D (7121)
- Apr 30, 2015-A (7121)
- Apr 30, 2015-B (7121)
- May 02, 2015-A (7121)
- May 02, 2015-B (7121)
- May 03, 2015-A (7121)
- May 03, 2015-A (7234)
- May 04, 2015-A (7234)
- May 05, 2015-A (7234)
- May 06, 2015-A (7234)
- May 06, 2015-B (7234)
- May 07, 2015-A (7234)
- May 08, 2015-A (7234)
- May 08, 2015-B (7234)
- Nov 02, 2014-A (7121)
- Nov 03, 2014-A (7121)
- Nov 03, 2014-B (7121)
- Nov 03, 2014-C (7121)
- Nov 03, 2014-D (7121)
- Nov 03, 2014-E (7121)
- Nov 04, 2014-A (7121)
- Nov 04, 2014-B (7121)
- Nov 04, 2014-C (7121)
- Nov 11, 2014-A (7121)
- Nov 11, 2014-B (7121)

3. Processing Summary

3.1. Flight Logs

Flight logs were completed by LIDAR sensor technicians for each mission during acquisition. These logs depict a variety of information, including:

- Job / Project #
- Flight Date / Lift Number
- FOV (Field of View)
- Scan Rate (HZ)
- Pulse Rate Frequency (Hz)
- Ground Speed
- Altitude
- Base Station
- PDOP avoidance times
- Flight Line #
- Flight Line Start and Stop Times
- Flight Line Altitude (AMSL)
- Heading
- Speed
- Returns
- Crab

Notes: (Visibility, winds, ride, weather, temperature, dew point, pressure, etc). Project specific flight logs for each sortie are available in Appendix A.

3.2. LiDAR Processing

Applanix + POSPac Mobile Mapping Suite software was used for post-processing of airborne GPS and inertial data (IMU), which is critical to the positioning and orientation of the LiDAR sensor during all flights. POSPac combines aircraft raw trajectory data with stationary GPS base station data yielding a "Smoothed Best Estimate Trajectory (SBET) necessary for additional post processing software to develop the resulting geo-referenced point cloud from the LiDAR missions.

During the sensor trajectory processing (combining GPS & IMU datasets) certain statistical graphs and tables are generated within the Applanix POSPac processing environment which are commonly used as indicators of processing stability and accuracy. This data for analysis include: Max horizontal / vertical GPS variance, separation plot, altitude plot, PDOP plot, base station baseline length, processing mode, number of satellite vehicles, and mission trajectory. All relevant graphs produced in the POSPac processing environment for each sortie during the Photo Science project mobilization are available in Appendix A.

The generated point cloud is the mathematical three dimensional composite of all returns from all laser pulses as determined from the aerial mission. Laser point data are imported into TerraScan and a manual calibration is performed to assess the system offsets for pitch, roll, heading and scale. At this point this data is ready for analysis, classification, and filtering to generate a bare earth surface model in which the above-ground features are removed from the data set. Point clouds were created using the Leica ALS Post Processor software. GeoCue distributive processing software was used in the creation of some files needed in downstream processing, as well as in the tiling of the dataset into more manageable file sizes. TerraScan and TerraModeler software packages were then used for the automated data classification, manual cleanup, and bare earth generation. Project specific macros were developed to classify the ground and remove side overlap between parallel flight lines.

All data will manually be reviewed and any remaining artifacts removed using functionality provided by TerraScan and TerraModeler. Global Mapper will be used as a final check of the bare earth dataset. GeoCue was used to create the deliverable industry-standard LAS files for both the All Point Cloud Data and the Bare Earth. In-house software will then used to perform final statistical analysis of the classes in the LAS files.

Metadata was generated for the project on a project, deliverable, and lift level.

3.3. LAS Classification Scheme

The classification classes are determined by the USGS Version 1.0 specifications and are an industry standard for the classification of LIDAR point clouds. All data starts the process as Class 1 (Unclassified), and then through automated classification routines, the classifications are determined using TerraScan macro processing.

The classes used in the dataset are as follows and have the following descriptions:

- Class 1 – Processed, but Unclassified – These points would be the catch all for points that do not fit any of the other deliverable classes. This would cover features such as vegetation, cars, etc.
- Class 2 – Bare earth ground – This is the bare earth surface
- Class 7 – Low Points – Low points, manually identified above or below the surface that could be noise points in point cloud.
- Class 9 – In-land Water – Points found inside of inland lake/ponds
- Class 10 – Ignored Ground – Points found to be close to breakline features. Points are moved to this class from the Class 2 dataset. This class is ignored during the DEM creation process in order to provide smooth transition between the ground surface and hydro flattened surface.
- Class 17 – Bridge Decks – Points falling on bridge decks.
- Class 18 – High Noise – High points, manually identified above or below the surface that could be noise points in point cloud.
- Class 25 – Overlap Water – Points found in the overlap between flight lines that are located inside hydro features. These points are created through automated processing methods and not cleaned up during processing.

3.4. Classified LAS Processing

The bare earth surface is then manually reviewed to ensure correct classification on the Class 2 (Ground) points. After the bare- earth surface is finalized; it is then used to generate all hydro-breaklines through heads-up digitization.

All ground (ASPRS Class 2) LiDAR data inside of the Lake Pond and Double Line Drain hydro flattening breaklines were then classified to water (ASPRS Class 9) using TerraScan macro functionality. A buffer of 3 feet was also used around each hydro flattened feature to classify these ground (ASPRS Class 2) points to Ignored ground (ASPRS Class 10). All Lake Pond Island and Double Line Drain Island features were checked to ensure that the ground (ASPRS Class 2) points were reclassified to the correct classification after the automated classification was completed.

All overlap data was processed through automated functionality provided by TerraScan to classify the overlapping flight line data to approved classes by USGS. The overlap data was classified through automated processes only and was not verified for classification accuracy. Due to software limitations within TerraScan, these classes were used to trip the withheld bit within various software packages. These processes were reviewed and accepted by USGS through numerous conference calls and pilot study areas.

All data was manually reviewed and any remaining artifacts removed using functionality provided by TerraScan and TerraModeler. Global Mapper is used as a final check of the bare earth dataset. GeoCue was then used to create the deliverable industry-standard LAS files for all point cloud data. Quantum Spatial proprietary software was used to perform final statistical analysis of the classes in the LAS files, on a per tile level to verify final classification metrics and full LAS header information.

3.5. Hydro-Flattening Breakline Process

Class 2 LiDAR was used to create a bare earth surface model. The surface model was then used to heads-up digitize 2D breaklines of Inland Streams and Rivers with a 100 foot nominal width and Inland Ponds and Lakes of 2 acres or greater surface area.

Elevation values were assigned to all Inland Ponds and Lakes, Inland Pond and Lake Islands, Inland Streams and Rivers and Inland Stream and River Islands using TerraModeler functionality.

Elevation values were assigned to all Inland streams and rivers using Quantum Spatial proprietary software.

All ground (ASPRS Class 2) LiDAR data inside of the collected inland breaklines were then classified to water (ASPRS Class 9) using TerraScan macro functionality. A buffer of 3 feet was also used around each hydro flattened feature. These points were moved from ground (ASPRS Class 2) to Ignored Ground (ASPRS Class 10).

The breakline files were then translated to ESRI Shapefile format using ESRI conversion tools.

3.6. Hydro-Flattening Raster DEM Process

Class 2 LiDAR in conjunction with the hydro breaklines were used to create a 1-meter Raster DEM. Using automated scripting routines within ArcMap, an ERDAS Imagine IMG file was created for each tile. Each surface is reviewed using Global Mapper to check for any surface anomalies or incorrect elevations found within the surface.

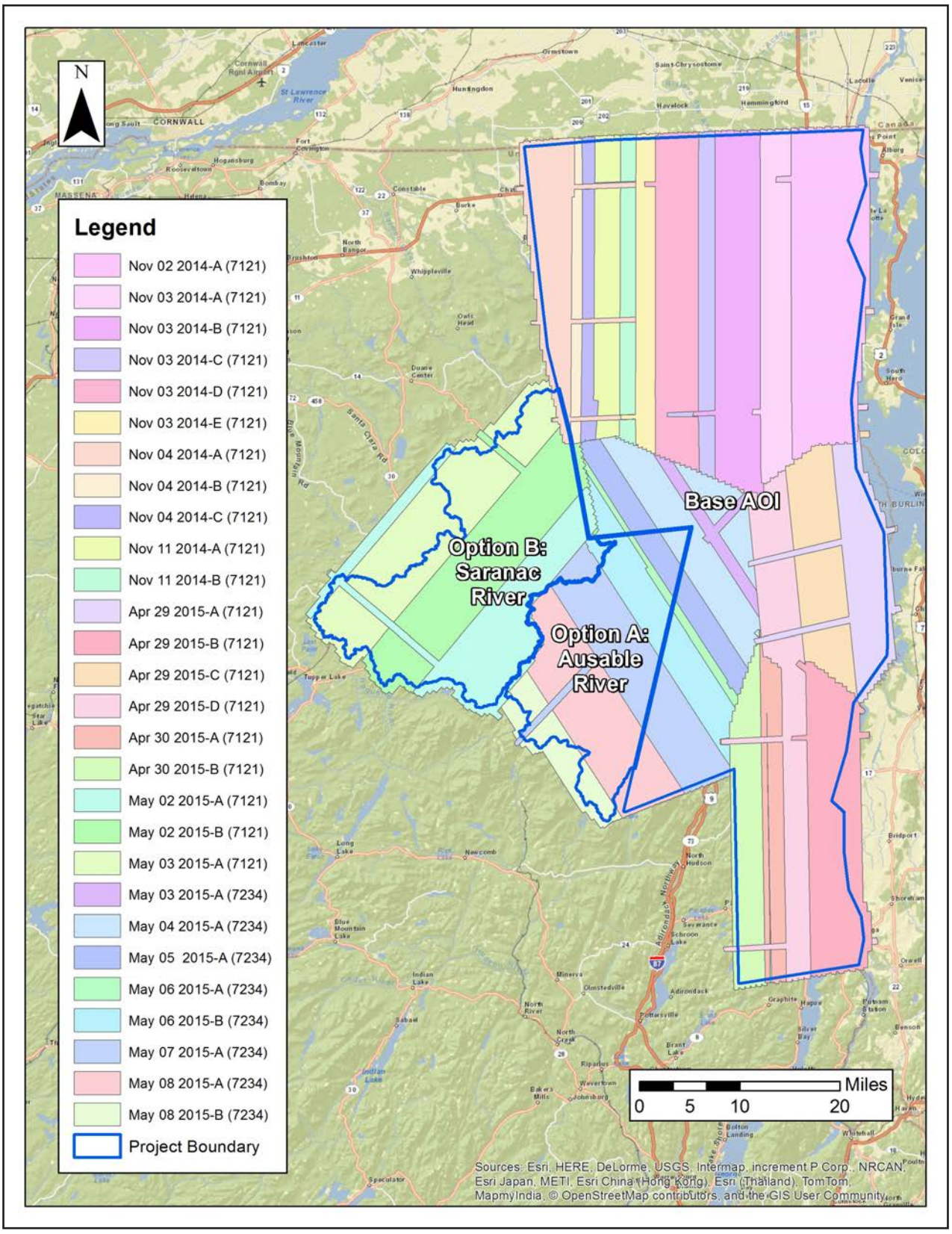
3.7 Contour Creation Process

Using automated scripting routines within ArcMap, a terrain surface was created using the ground (ASPRS Class 2) LiDAR data as well as the hydro-flattening breaklines. This surface was then used to generate the final 1 meter contour dataset in ESRI shapefile format.

4. Project Coverage Verification

Coverage verification was performed by comparing coverage of processed .LAS files captured during project collection to generate project shape files depicting boundaries of specified project areas. Please refer to Figure 5.

Figure 5. Flightline Swath LAS File Coverage



5. Ground Control and Check Point Collection

Quantum Spatial completed a field survey of 76 ground control (calibration) points along with 104 QA points in 4 land cover classifications (total of 180 points) as an independent test of the accuracy of this project.

A combination of precise GPS surveying methods, including static and RTK observations were used to establish the 3D position of ground calibration points and QA points for the point classes above. GPS was not an appropriate methodology for surveying in the forested areas during the leaf-on conditions for the actual field survey (which was accomplished after the LiDAR acquisition). Therefore the 3D positions for the forested points were acquired using a GPS-derived offset point located out in the open near the forested area, and using precise offset surveying techniques to derive the 3D position of the forested point from the open control point. The explicit goal for these surveys was to develop 3D positions that were three times greater than the accuracy requirement for the elevation surface. In this case of the QA points the goal was a positional accuracy of 5 cm in terms of the RMSE.

Figure 6 shows the location of each bare earth calibration point for the project area. Table 4 depicts the Control Report for the LiDAR bare earth calibration points, as computed in TerraScan as a quality assurance check. Note that these results of the surface calibration are not an independent assessment of the accuracy of these project deliverables, but the statistical results do provide additional feedback as to the overall quality of the elevation surface.

The project was delivered using the following horizontal projection(s): NAD83 UTM Zone 18, Meters; MAVD88 (Geoid 12A), Meters. In this document, horizontal coordinates for ground control and QA points for all LiDAR classes are reported in the projection stated above.

The required accuracy testing was performed on the LiDAR dataset (both the LiDAR point cloud and derived DEM's) according to the USGS LiDAR Base Specification Version 1.0. The summary below provides the results of this testing.

5.1. Point Cloud Testing

Raw Fundamental Vertical Accuracy (Raw FVA): The tested Raw FVA for the dataset was found to be 0.066 meters in terms of the RMSEz. The resulting FVA stated as the 95% confidence level (RMSEz x 1.96) is 0.129 meters. This dataset meets the required FVA of 18.13 cm at the 95% confidence level (according to the National Standard for Spatial Database Accuracy (NSSDA)), based on TINs derived from the final calibrated and controlled LiDAR swath data. See Figure 6.

5.2. Digital Elevation Model (DEM) Testing

Fundamental Vertical Accuracy (FVA)

The tested FVA for the dataset captured from the DEM using bi-linear interpolation to derive the DEM elevations was found to be 0.066 meters for Lot 4 and 0.058 meters for Lot 5 in terms of the RMSEz. The resulting accuracy stated as the 95% confidence level (RMSEz x 1.96) is 0.130 meters and 0.114 meters for Lot 4 and 5, respectively. This dataset meets the required FVA of 18.13 cm at the 95% confidence level (based on NSSDA). See Figure 7.

Supplemental Vertical Accuracy (SVA)

The tested SVA accuracies for the dataset for each of the land cover classes other than open ground are summarized below. These results are stated in terms of the 95th percentile error (based on ASPRS guidelines) for each of the land cover classes other than open ground.

The following land cover classes were tested and the resulting 95th percentile error values are listed below:

- Forested, Fully Covered by Trees: 0.140 meters (Figure 8)
- Swamp/Wetland: 0.315 meters (Figure 9)
- Tall Weeds: 0.266 meters (Figure 10)

Consolidated Vertical Accuracy (CVA)

The tested CVA for the dataset captured from the DEM using bi-linear interpolation for all classes (including the bare earth class) was found to be 0.228 meters, which is stated in terms of the 95th percentile error. Therefore the data meets the required CVA of 26.9 cm. This test was based on the 95th percentile error (based on ASPRS guidelines) across all land cover categories.

Figure 6. Ground Control Points Used in Calibration

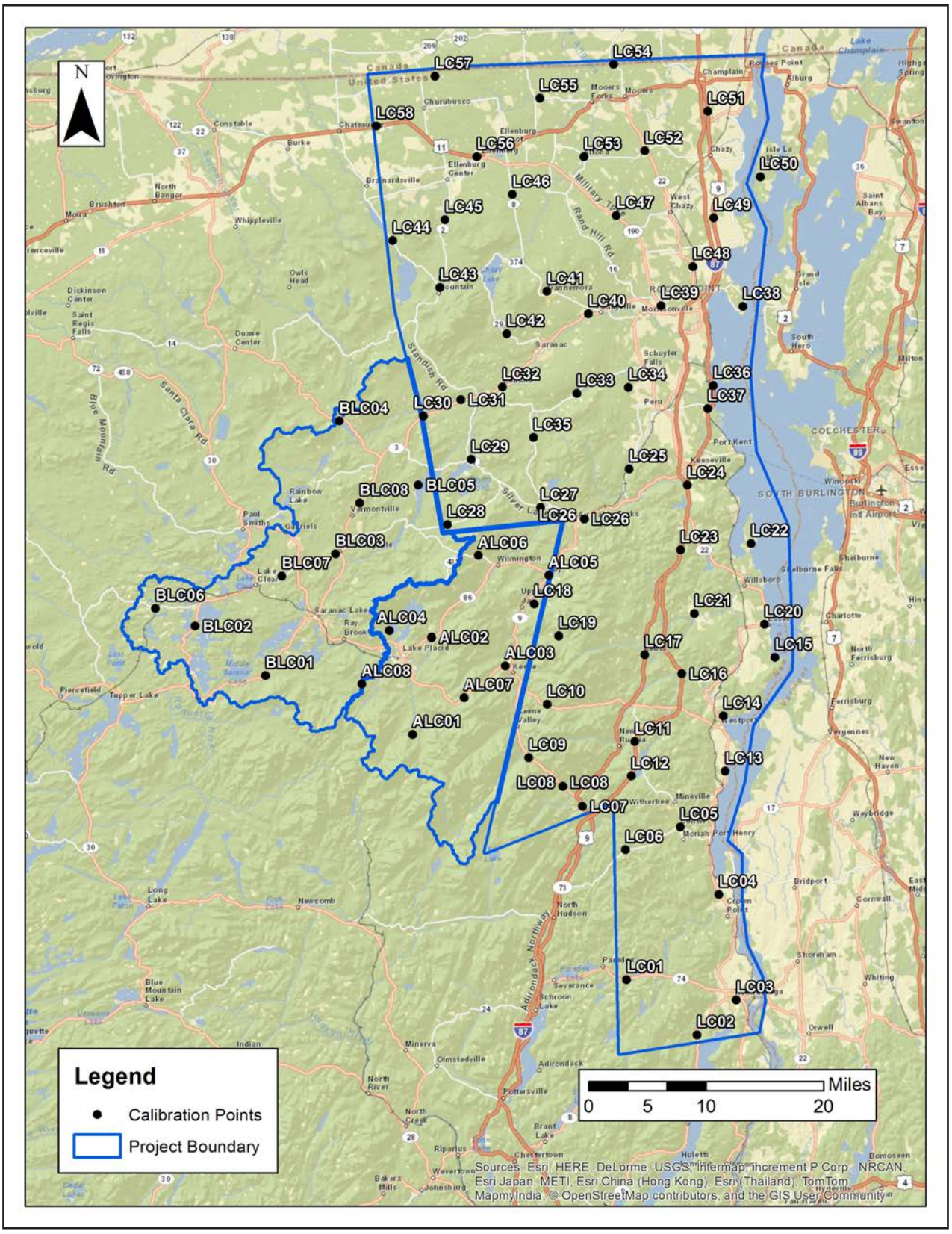


Figure 7. Bare Earth QA Point Locations

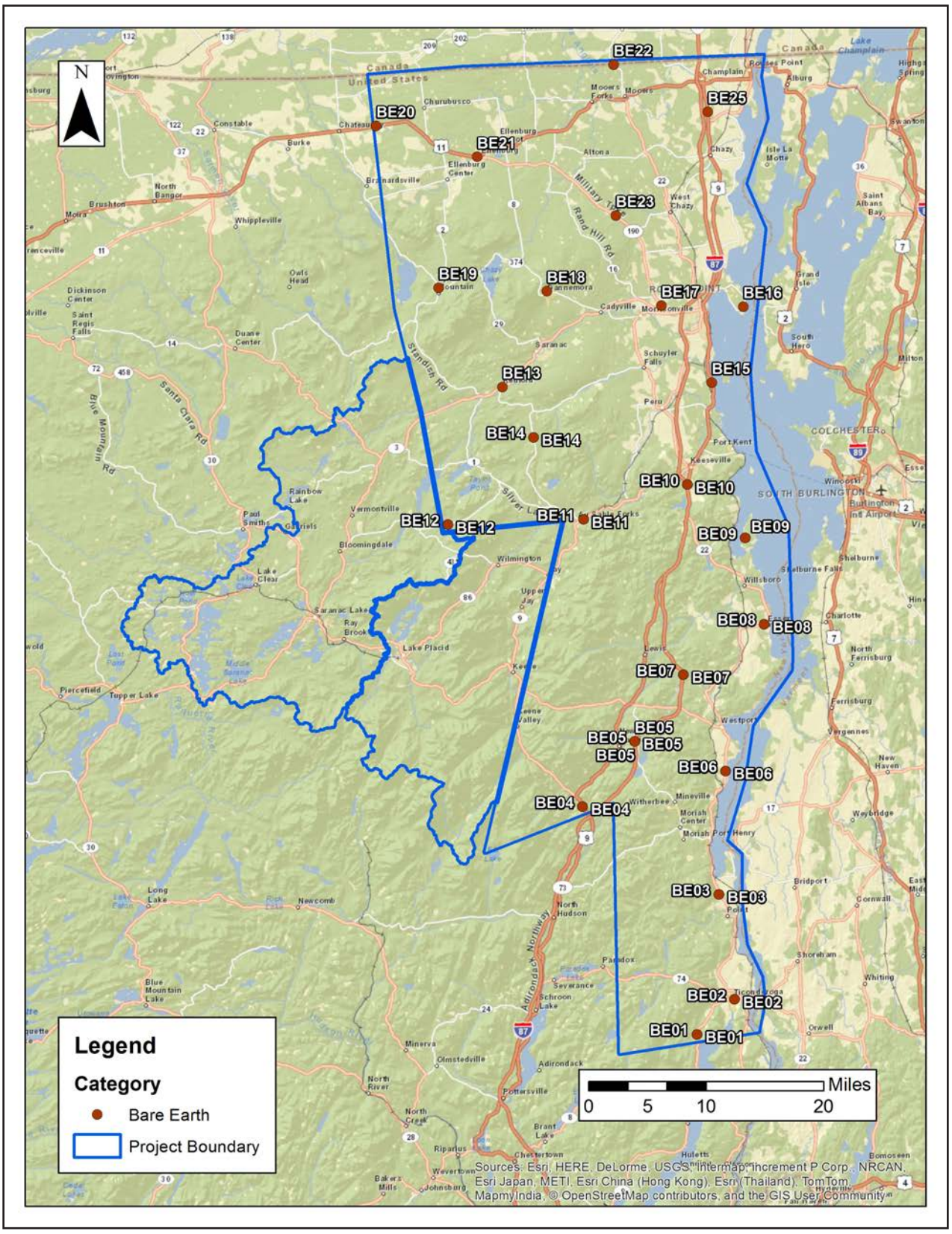


Figure 8. Forested QA Point Locations

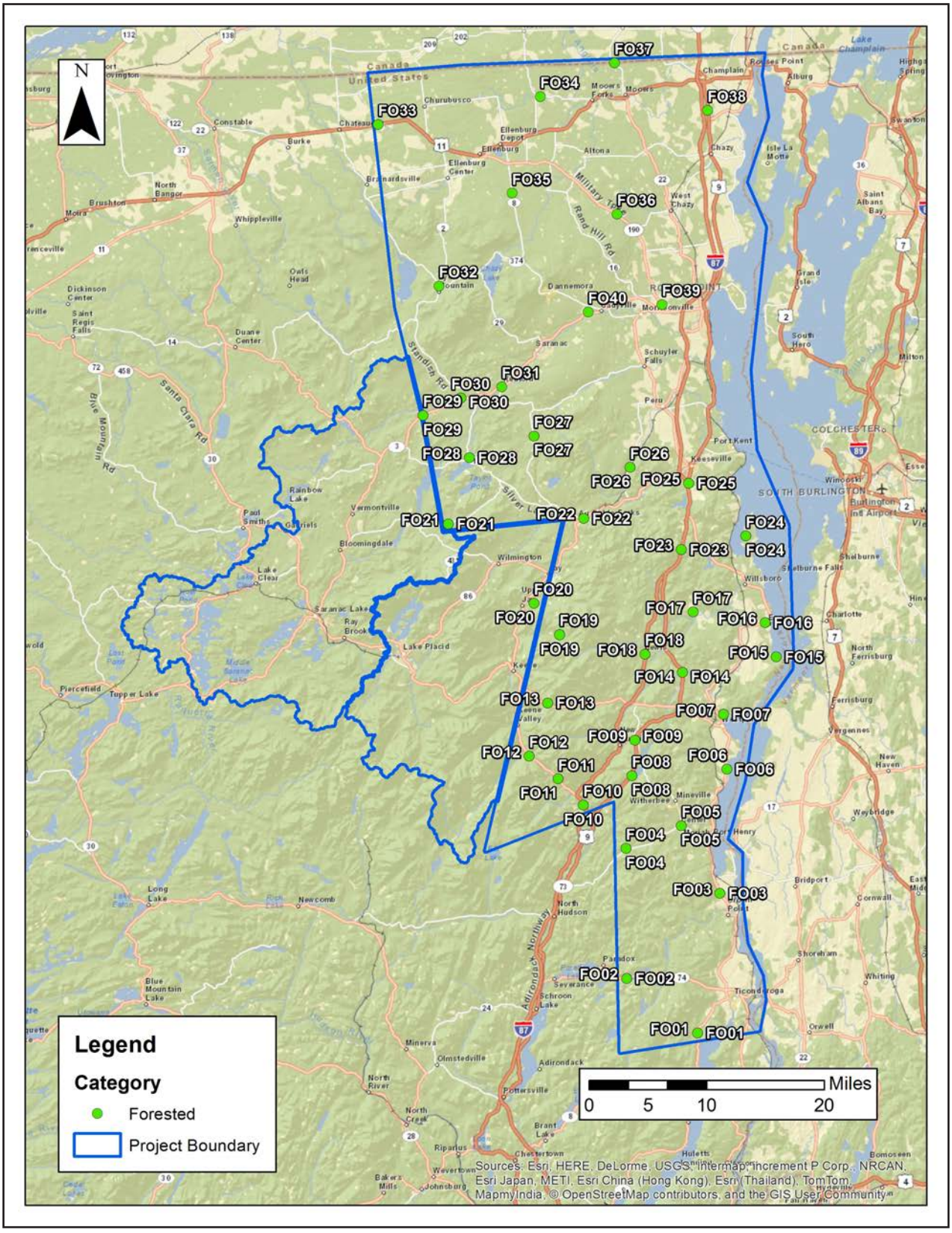


Figure 9. Swamp/Wetland QA Point Locations

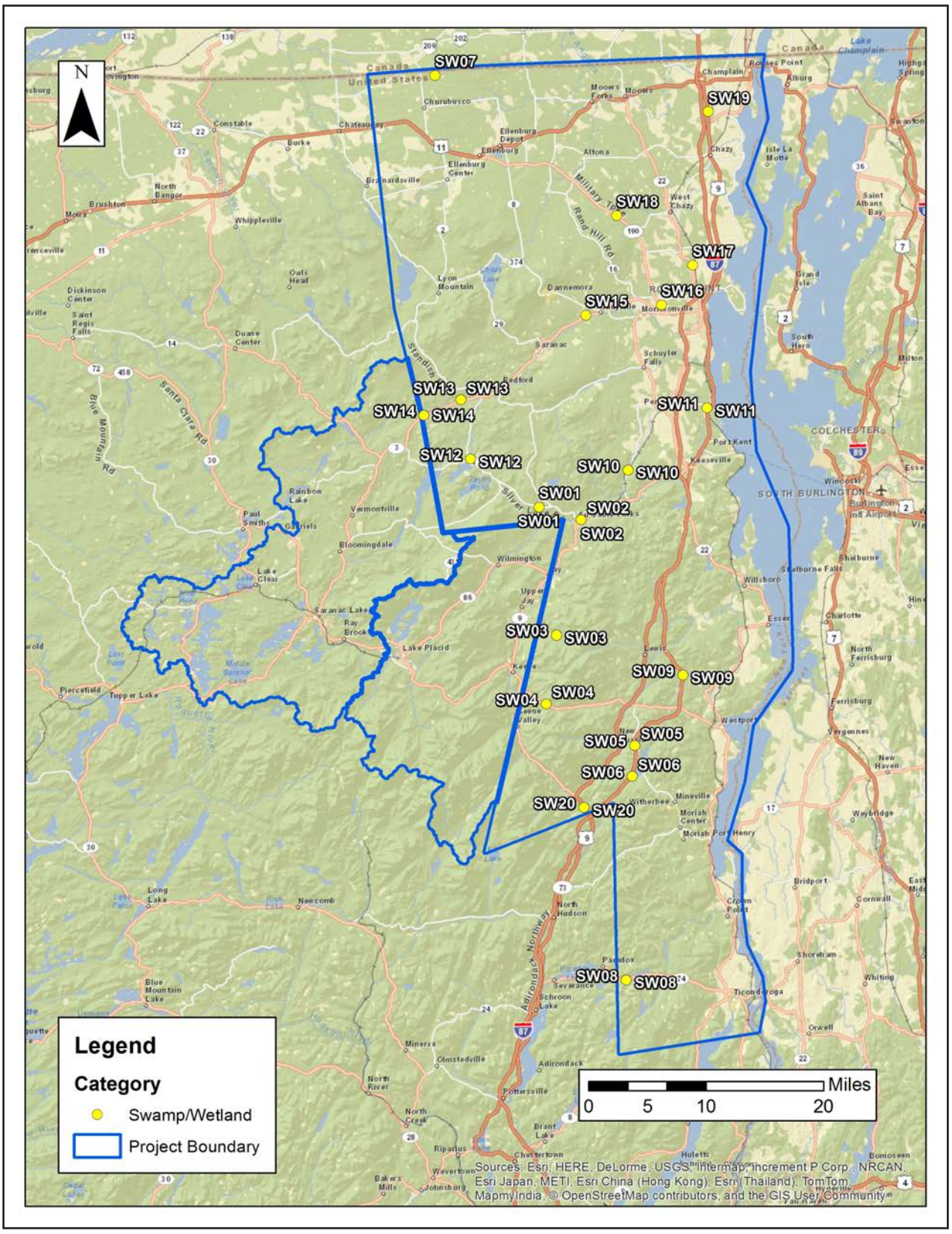


Figure 10. Tall Weeds QA Point Locations

